



Burnout: Redefining its key symptoms

Gabriela Tavella, Dusan Hadzi-Pavlovic, Gordon Parker^{*}

School of Psychiatry, University of New South Wales, Sydney, NSW, Australia

ARTICLE INFO

Keywords:
Stress
Depression
Diagnosis
Burnout

ABSTRACT

Burnout is currently operationalised as comprising of exhaustion, depersonalization and reduced personal accomplishment. However, questions have been raised as to whether this three-factor model accurately characterises the syndrome. We therefore sought to re-examine burnout's key symptoms and to develop a new model of the syndrome. 622 participants who self-identified as experiencing burnout completed a questionnaire covering a large item set of candidate burnout symptoms. Bifactor modelling identified a 34-item general factor that was dominated by items capturing exhaustion and cognitive dysfunction, but which also included items indicative of decreased work performance, insularity and a depressed mood. Five specific factors capturing additional data variance were identified and were interpreted as representing cognitive impairment, empathy loss, exhaustion, compromised work performance, and social withdrawal. Reliability indices indicated that the general factor alone accounted for most of the variance in observed scale scores for each specific factor, except for the loss of empathy specific factor. We concluded that burnout may therefore potentially be modelled as a unidimensional construct comprising exhaustion, cognitive impairment, compromised work performance, empathy loss and social withdrawal. Further, while those with burnout are likely to experience depressive symptoms, their presence is not of necessity indicative of clinical depression.

Introduction

Burnout is a topic of high public and research interest reflecting its severe consequences, with costs to the global economy estimated at over \$300 billion annually and a burnout “pandemic” forecast by the World Health Organisation (WHO) in the next decade (Bretland and Thorsteinsson, 2015). However, scientists and practitioners debate the precise definition of burnout, its key symptoms, and whether it is a distinct mental health condition (Bianchi et al., 2019; Cox et al., 2005; Heinemann and Heinemann, 2017). Burnout lacks formal diagnostic criteria – not being included in DSM-5 (APA, 2013) and listed in the ICD-11 “not as a medical condition” (WHO, 2019) but as a “factor influencing health status” (WHO, 2018). Despite such issues, for the general public “there seems to be no doubt that burnout is a real and serious phenomenon” (Heinemann and Heinemann, 2017), and the syndrome is a diagnosable condition in some European countries, with a diagnosis making individuals eligible for financially compensated sick leave and insurance benefits (Schaufeli et al., 2009). Burnout's diagnostic status is therefore clearly ambiguous, with the need to determine how best to define and measure the syndrome imperative to evaluating its status as a distinct nosological category. We report here a study designed to re-examine

burnout's key features and which offers a new operational definition of the syndrome.

The Maslach Burnout Inventory (MBI; Maslach et al., 2016) is the most common tool used to define and measure burnout and its triadic model underpins the ICD-11 definition. It was initially developed by Maslach and Jackson (1981), whose exploratory interviews with human services staff led them to predict burnout to be a syndrome comprising (i) exhaustion, (ii) cynicism or loss of empathy, and (iii) reduced professional accomplishment. They created a 47-item questionnaire of such symptoms that was completed by human service workers. The item set was reduced via a series of factor analyses, with the final 22-item measure consisting of an “emotional exhaustion” subscale, a “depersonalization” subscale capturing empathy loss, and a “personal accomplishment” subscale encompassing decreased work performance. They later adapted the original MBI (the MBI-Human Services Survey or MBI-HSS) for use in different occupations, with one such adaptation being the MBI-General Survey (MBI-GS) designed to measure burnout experienced by those working outside of the human services sector (Maslach et al., 2016).

The MBI was not designed to be used as a categorical tool for determining burnout “caseness”. Rather, the MBI positions burnout as a

^{*} Corresponding author at: School of Psychiatry, Level 1 AGSM Building, UNSW, Sydney 2052, Australia.

E-mail address: g.parker@unsw.edu.au (G. Parker).

dimensional construct varying in severity, with the authors arguing that there is “no definitive score that ‘proves’ a person is ‘burned out’” (Maslach et al., 2016). Thus, the MBI’s burnout model should not be considered as diagnostic, and while some researchers have since imposed various cut-off scores for diagnostic purposes, these have been criticised as arbitrary and inconsistent across studies (Bianchi et al., 2018). As the MBI is not diagnostic, it should also not be considered as having differentially diagnostic status, especially as items assessing psychological symptoms (e.g. anxiety, depression) beyond exhaustion, empathy loss and reduced professional accomplishment, were not, to our knowledge, evaluated for inclusion in the MBI’s burnout definition. The MBI manual (Maslach et al., 2016) does discuss burnout’s distinction from several other psychological constructs including job dissatisfaction, job stress, and depression. In relation to the latter, the authors’ position burnout as independent from depression, citing studies (e.g., Leiter and Durup, 1994) showing that MBI items load onto different factors than depression items in factor analytic solutions. However, several other studies show significant overlap between burnout and depression symptoms (e.g., Aloha et al., 2005; Dyrbye et al., 2008) and high correlations between the two conditions (e.g., Bianchi et al., 2013; Glass & McKnight, 1996). Thus, while the MBI’s burnout definition does not include any depressive symptoms, this is possibly only a consequence of such symptoms not being included in the initial MBI questionnaire, and thus whether depressive symptoms are integral to the definition of burnout (or burnout is synonymous with depression) remains to be conclusively determined.

As the MBI was designed specifically to measure Maslach and Jackson’s hypothesised three-factor definition of burnout, it has been criticised for being derived through factor analysing only a small set of “arbitrary” items representing these constructs (Schaufeli, 2003). It is therefore possible that the MBI’s final burnout definition would have differed if a more exhaustive list of items had been included in the analysis, as other dimensions of burnout may have been identified (Bianchi et al., 2015a). In an extensive review examining how burnout research has been conducted up until 2011, Heinemann and Heinemann (2017) concluded that “to date, most burnout research has been circular, because it relies on questionnaires that measure symptoms that have not (yet) been clearly defined or unanimously agreed upon in medicine and psychology” and that any newer research attempting to overcome such issues is “equally problematic, because it uses the same debatable and contested constructs and inventories to identify participants who purportedly suffer from burnout, and then examines this mental condition in these individuals.” They argued that future burnout research should focus on determining an accurate burnout definition not based on previously established burnout measures, and that any new research should focus on elucidating whether burnout can be distinguished from other psychological conditions, such as depression.

Considering such arguments, we initiated an exploratory study (Tavella et al., 2020) to re-examine burnout’s key features and how it should best be operationalised. Study participants completed a 106-item questionnaire capturing candidate burnout symptoms and some personality descriptors. Factor analyses allowing correlated factors of the data were undertaken. In all solutions generated by the exploratory factor analyses, the first factor (unrotated) accounted for over 30% of variance, which was judged as potentially suggesting a *general factor* accounting for most of the variance in the dataset, so we therefore undertook a bifactor analysis of the data. The bifactor solution with a general factor (representing variance in the data accounted for by the ‘primary’ or ‘target’ dimension being measured) and three additional ‘specific’ factors (which account for item response variance not captured by the general factor) was judged as the solution most adequately capturing the data. The general factor had items weighting exhaustion, cognitive problems, social withdrawal, as well as worry and depressive symptoms. The first specific factor comprised empathy loss in conjunction with social withdrawal, which we interpreted as a general “inability to feel” factor. The second factor encompassed cognitive dysfunction

coupled with reduced work efficacy, while the third captured a perfectionistic personality style. Thus, our analysis showed some overlap with the MBI, but also several differences, most notably in identifying a compromised cognition component and allowing a diathesis-stress model with burnout being a consequence of work stressors and a predisposing personality style of perfectionism.

Three key questions arose from that study. Firstly, would the symptom clusters identified in that study be prevalent in a new sample of participants self-identifying as burnt out? Secondly, is burnout best defined by a single, overarching, symptom dimension (as suggested by our general factor in the first study), and therefore potentially capable of being measured by a single scale, or are there independent symptom dimensions (represented by the specific factors) arguing for additional subscales in any measure? Thirdly, is burnout independent of depression? We now report a second study seeking such clarification.

Methods

2.1. Participants

Individuals could participate if they were fluent in English, aged 18–65, and self-identified as currently experiencing ‘burnout,’ whether involved in formal/paid work, informal/unpaid work, or retired. Participants were recruited through advertisements on the Black Dog Institute website.

2.2. Procedure

Participants provided online consent and then completed an anonymous online questionnaire that first obtained demographic and employment data. The questionnaire then listed 137 potential symptoms of burnout, with participants required to judge whether they experienced each distinctly, moderately, slightly, or not at all (coded 3, 2, 1, and 0, respectively). 106 of these items were included in our first study (Tavella et al. 2020), and were derived from previous burnout measures, a detailed review of the burnout literature—which allowed for the inclusion of features nominated in the literature but not included in many burnout measures, such as symptoms of cognitive dysfunction (Beck et al., 2013; Schaufeli and Taris, 2005; Sandström et al. 2005)—and the clinical experience of the third author (GP). Several items were included to capture features of each MBI-HSS and MBI-GS subscale (but were reworded to avoid plagiarism), for instance, “I wake up feeling tired” was included to represent “I feel tired when I wake up in the morning and have to face another day on the job” from the exhaustion subscale of the MBI-GS (Maslach, et al., 2016). Items representing those from the personal accomplishment/professional efficacy subscales of the MBI-HSS and MBI-GS were included but negatively valenced in our list, to remain consistent in direction with other items. For instance, “I can easily understand how my recipients feel about things” from the MBI-HSS (Maslach et al., 2016) was replaced with “I struggle to understand the feelings of colleagues, customers and/or recipients of my care” in our scale.

In addition to items from our first study, we collated items from four validated depression measures - the Patient Health Questionnaire (PHQ-9; Kroenke and Spitzer, 2002) with its items covering most of the DSM-5 criteria of Major Depressive Disorder); the Hamilton Depression Rating Scale (HDRS; Hamilton, 1960); the Montgomery Åsberg Depression Rating Scale-Self-assessment (MADRS-S; Svanborg and Åsberg, 1994); and the Beck Depression Inventory (BDI; Beck et al., 1961). Several items from across these scales covered very similar symptoms, such as “a gloomy attitude/feelings of sadness” (HDRS) and “Feeling down, depressed, or hopeless” (PHQ-9), and were therefore replaced by a single item in our list (e.g., “I feel sad, empty and hopeless”). We also included items from the Sydney Melancholic Prototypic Index (SMPI; Parker et al., 2013) that capture melancholic and non-melancholic depressive symptoms, as well as additional criteria not covered by the

depression measures, such as feelings of “leaden paralysis” which is included in the DSM-5 as a specifier for atypical (non-melancholic) depression. A total of 31 depression items were included in our final list. Participants were also asked later in the questionnaire if they had ever been diagnosed with depression by a doctor or mental health professional.

Other sections of the questionnaire sought to assess burnout triggers, the impact of burnout, whether participants had sought help for their burnout, and personality style, but such data are not detailed here as this paper considers our analyses of burnout symptoms only.

2.3. Statistical analyses

Bifactor and non-bifactor factor analyses were carried out in Mplus (Muthén and Muthén, 2002), with data treated as categorical and using the weighted least squares mean and variance (WLSMV) estimator. As detailed in the Introduction, bifactor modelling was judged as most accurately capturing the data in our first study (Tavella et al., 2020). We also prioritised exploratory bifactor analyses in this study, as several studies report moderate to high correlations between the three dimensions of the MBI's burnout construct (see Worley et al., 2008) and such correlations suggest a hierarchical factor structure (Mészáros et al., 2014). Bifactor modelling has been put forward as a “more effective approach to modelling construct-relevant multidimensionality” than commonly-used second-order models (Mészáros et al., 2014; see also Brunner et al., 2012 and Chen et al., 2006). As noted, bifactor analyses load all items onto a single general factor capturing the primary domain being measured (‘overall’ burnout in this instance), while also specifying one or more secondary specific factors that are uncorrelated with the general factor (and in the models we examined, uncorrelated with each other – the standard definition) and accounting for any additional variance in the data set not captured by the general factor.

The item set was repeatedly reduced between sequential EFAs. Initial candidates for deletion were obvious duplicate or similar items and ones making a minimal contribution to the solution (i.e. loading less than approximately 0.35 on any factor). For our progressively refined 61-item and 43-item solutions we also calculated for each item (i) item characteristic curves (ICCs), which show the probability of someone responding to each of the response categories (from “not at all” to “distinctive”) as a function of their level on the underlying factor (Asparouhov and Muthén, 2020), and (ii) a degree of severity (D) value, which corresponds to the point on the latent dimension where a response of “moderate” becomes more likely than a response of “slight”, with a broad range of D-values across items increasing the likelihood that a scale based on those items can discriminate between different levels of the latent construct. Pairs of items with similar content, loadings and D-values were judged as indicating overlap and redundancy, and thus the item of the pair with the inferior ICC was deleted. Item redundancy was also indicated in pairs of items with similar content, loadings and ICCs, and so the item from such pairs whose deletion would result in evening up the distribution of D-values across the solution was selected for deletion.

As noted by Reise et al. (2016), model fit statistics cannot be used to choose between a bifactor model and a correlated factor model. However, some may find such statistics informative when considering the fit of specific models, and we therefore calculated comparative fit (CFI), Tucker–Lewis (TLI), and root mean square error approximation (RMSEA) indices for our final model. CFI and TLI indices larger than 0.90 have been reported as indicative of adequate model fit (e.g., Garson, 2015; Hu & Bentler, 1999), while a RMSEA index between 0.08 and 0.05 suggests adequate fit (Wan, 2002; Hu & Bentler, 1999) and a RMSEA below 0.05 is indicative of an excellent fit (Garson, 2015).

We calculated reliability estimates for each factor in our final model by calculating coefficient ω and hierarchical coefficient ω_H indices for each factor (Rodríguez et al., 2016). These statistics are estimates of how precise scale scores from each factor are in measuring the relevant

construct (Mészáros et al., 2014). Coefficient ω represents the proportion of variance in scale scores from each factor that is accounted for by a blend of both the general and specific factors. Coefficient ω_H is an estimate of the variance in scale scores accounted for by the relevant factor only (i.e., after “partitioning out” the variance for the general factor; Rodríguez et al., 2016). As there is no generally accepted calculation of these indices when items are categorical and the WLSMV estimator is used, reliability statistics were based on treating the items as continuous.

A risk of relying on self-identification of burnout by participants is that a percentage may have had a primary psychological condition other than burnout, particularly depression. We therefore calculated and compared fit statistics for subsets of participants who reported that they had or had not been previously diagnosed with depression to examine for measurement invariance. In addition, we fitted a multiple-group model using these subsets, with constraints defined for scalar invariance—where all factor means were set to zero for the subset previously diagnosed with depression and set free in the other subset—and tested if the factor means in the latter subset differed from zero.

Results

3.1. Demographics

Data from 622 completed questionnaires were analysed. The sample was predominantly female (78.4%) with a mean age of 41.5 years. Most identified as non-Indigenous Australians (71.9%), had undergraduate (36.3%) or postgraduate (41.2%) degrees and were employed full-time (69.0%) or part-time (17.0%). The five most commonly nominated occupations were education professionals (e.g. teacher, lecturer; 10.6%), specialty managers (e.g. advertising manager, sales manager; 10.0%), public servants (8.0%), community/welfare workers (e.g., social workers; 7.2%), and midwives/nurses (5.6%).

3.2. Factor analyses

Exploratory bifactor analyses of the 137-item set were undertaken, with bifactor (orthogonal Geomin) rotations with a general factor plus one to nine specific factors generated (as solutions with more than nine specific factors had multiple factors with few items with loadings > 0.35). The solution with one general factor plus six specific factors (hereafter referred to as the 137-item GF + 6SF solution) was judged as the most informative, with its general factor including a broad range of highly loading items (including depression and anxiety items), while its six specific factors captured (i) empathy loss, (ii) reduced work performance, (iii) energy depletion, (iv) sleep disturbance, (v) cognitive dysfunction, and (vi) social withdrawal/disconnection, with the first three in accord with MBI constructs, and the latter three capturing constructs identified in our earlier study (Tavella et al., 2020; Tavella and Parker, 2020a).

We compared the bifactor analysis against a factor analysis with correlated factors (i.e. non-bifactor, oblique Geomin rotation) of the 137 item-set. Approximately one-third of the items loaded on to the first factor in the correlated factors solutions, suggesting that a single (i.e. ‘general’) factor accounted for the majority of variance. We therefore judged bifactor analysis as appropriate for further refined analyses.

Obvious duplicate items (e.g., “I cannot look forward to things that would normally give me pleasure” and “I look forward to pleasurable things less”) and ones making a minimal contribution to the solution were removed, with items retained being all those with loadings greater than 0.35 on specific factors, as well as all additional items loading highly on the general factor (i.e. loadings > 0.7) but not loading on any of the specific factors, with 61 items remaining and with their loadings on the general factor ranging from 0.19 to 0.77.

We ran bifactor analyses of the 61 items. The solution with four specific factors (hereafter referred to as the 61-item GF + 4SF solution) was interpreted as best capturing the data as its general factor weighted

a broad range of highly loading exhaustion, cognitive and depressive items, and its specific factors covered all those domains captured in the 137-item GF + 6SF solution (albeit combining the energy depletion and sleep disturbance factors into a single exhaustion factor), except for social withdrawal - with items for that construct instead loading distinctly on the general factor.

Several highly loading items remained that captured similar descriptors (e.g. "I socialise less" and "I withdraw from family and friends"), indicating item redundancy and arguing for some item deletions. By comparing item content, loadings, ICCs, and D-values (as described previously), 18 items were deleted, and with the remaining 43 items maintaining a reasonable balance across D-values.

We re-ran the analyses with the refined 43 items, generating solutions with two to five specific factors (as previously any additional specific factors contained fewer than four items with loadings > 0.35, with such additional factors therefore judged to be of low utility). The imposed solution that generated one general factor plus five additional factors (hereafter referred to as the 43-item GF + 5SF solution) was interpreted as capturing clear specific constructs (described shortly) in addition to the general factor. This solution included a general factor with most of the highest loading items being capturing cognitive dysfunction, but also including some exhaustion and reduced work performance items. Other items on the general factor encompassed reduced motivation, low mood, feelings of reduced self-worth, social withdrawal/disconnection as well as empathy loss. Based on item loadings, we interpreted the five specific factors as capturing (i) cognitive dysfunction, (ii) empathy loss, (iii) exhaustion, (iv) reduced work performance, and (v) social withdrawal in conjunction with low mood. Thus, our solution included all constructs from the MBI while also indicating compromised cognition, social withdrawal and low mood.

We ran an additional analysis to check that the use of bifactor as opposed to a factor analysis with correlated factors (ie., non-bifactor) adequately captured the data. A correlated factor analysis with six factors (with orthogonal rotation) of the 43 items was computed and was almost identical to the 43-item GF + 5SF solution (see supplementary material), as most items loaded moderately to highly on the first factor, suggesting a single (i.e., general) factor accounted for most of the variance in items. Further, the factor structure of the remaining factors was similar to our five specific factors in the bifactor analysis in terms of item loadings. We therefore judged our bifactor analysis as appropriately representing the data.

3.3. A refined 34-item measure

Measures with fewer items are preferable as assessment tools as they reduce the burden on completers (Bowling, 2005). We therefore sought to further reduce the item set while maintaining the basic structure of the 43-item GF + 5SF solution. All items in the solutions loaded at > 0.35 on at least one factor, and thus no items were removed due to low loadings. Candidate items for deletion were therefore identified using a similar process to that used to reduce the 61 item-set (i.e. using ICCs and D-values), resulting in a further nine items being deleted from the solution, and leaving a total of 34 items.

We re-ran the bifactor analysis with the 34 items to assess whether the factor structure of the 43-item GF + 5SF solution was maintained with fewer items. The bifactor rotation with a general factor and five specific factors (hereafter referred to as the 34-item GF + 5SF solution) is reported in Table 1, and we judged this solution as the preferable final one as it essentially retained the factor structure and loadings of the 43-item solution, while reducing the item-set by 21%.

As some readers may find fit statistics informative, we calculated such statistics for the 34-item GF + 5SF solution after items with loadings less than approximately 0.35 were set to zero (as per Table 1). These statistics are displayed in Table 2 and show adequate fit according to all indices.

We calculated reliability estimates for each of the factors in the 34-

Table 1

Item loadings for the 34-item GF + 5SF solution (items retained for specific factors were those with loadings of approximately >.35 and are bolded).

Item	GF	SF 1	SF 2	SF 3	SF 4	SF 5
I cannot concentrate or register new information because of foggy thinking	0.612	0.649	0.014	0.052	-0.060	0.028
I feel slowed down mentally (e.g. hard to find words, slowed thoughts)	0.604	0.610	-0.012	0.082	-0.024	-0.012
I have to re-read things because I was not concentrating the first time	0.565	0.486	-0.047	-0.011	0.083	-0.023
I find it hard to concentrate on the task at hand	0.729	0.446	-0.014	0.069	0.256	-0.112
I take longer to finish tasks at work	0.553	0.438	-0.069	0.005	0.428	0.001
My capacity to remember things is not as good as usual	0.689	0.427	-0.034	0.017	0.051	-0.009
My attention is less focussed	0.751	0.409	0.032	0.022	0.208	-0.079
I am less empathetic	0.248	-0.005	0.795	-0.042	-0.049	-0.006
I feel less empathy and sympathy towards people in general	0.466	-0.069	0.748	0.029	0.033	0.057
I struggle to understand the feelings of colleagues, customers and/or recipients of my care	0.310	0.059	0.622	-0.096	-0.037	0.025
I care less about what happens to people that I work with (e.g. colleagues, customers, recipient of my care)	0.463	-0.092	0.563	0.007	0.157	0.128
I constantly feel tired or fatigued	0.684	0.034	0.012	0.586	-0.034	0.015
I wake up feeling tired	0.731	0.009	-0.012	0.523	-0.101	-0.057
I am not refreshed by sleep	0.682	0.036	-0.036	0.507	-0.136	-0.018
I lack energy across the day	0.692	0.030	0.014	0.450	0.057	0.000
I feel worn out	0.738	0.033	-0.030	0.449	-0.006	0.070
I experience a loss of energy (making it hard to get going in the morning)	0.693	0.073	-0.041	0.445	0.049	-0.022
I am less productive at work	0.560	0.313	-0.032	0.059	0.677	0.004
The quality of my work output is lower	0.540	0.336	0.012	-0.068	0.633	-0.034
My work performance worsens	0.490	0.284	0.010	-0.077	0.576	0.022
I feel like I am making less of a difference at work	0.604	-0.122	0.055	-0.144	0.384	-0.087
I stop feeling very driven to keep meeting my responsibilities	0.586	-0.001	0.036	-0.012	0.376	-0.095
I cannot get pleasure out of my work	0.596	-0.147	0.133	0.020	0.351	-0.016
I stop looking forward to spending time with friends and family	0.572	-0.121	0.129	0.010	-0.019	0.669
I withdraw from family and friends	0.600	0.010	0.019	-0.005	-0.132	0.595
I keep to myself	0.589	0.003	0.034	-0.042	-0.086	0.457

(continued on next page)

Table 1 (continued)

I cannot look forward to things that would normally give me pleasure	0.688	-0.147	0.118	-0.001	-0.003	0.372
I feel emotionally drained and exhausted	0.788	-0.046	0.012	0.241	-0.012	0.088
I feel sad, empty and hopeless	0.768	-0.109	-0.111	-0.149	-0.068	0.143
I feel as though I am stagnating and life is passing me by	0.741	-0.176	-0.024	-0.180	0.057	-0.007
I start feeling more self-critical and hard on myself	0.733	0.019	-0.168	-0.198	-0.029	0.032
I find it more difficult to take life as it comes	0.728	-0.038	-0.012	-0.122	-0.091	0.068
I find little things and chores frustrating	0.683	-0.036	0.135	-0.024	-0.037	-0.049
I spend much of my days worrying	0.669	-0.016	-0.131	-0.143	-0.106	0.015

GF = general factor

SF = specific factor

Table 2

Model fit indices and difference testing for the 34-item GF + 5SF solution (when items not bolded in Table 1 were set to zero).

	Model Fit Indices					
	RMSEA	CFI	TLI	χ^2	df	p
Whole sample (n = 622)	0.064	0.936	0.928	1761.6	499	<0.0001
Previously diagnosed with depression (n = 333)	0.057	0.942	0.934	1047.8	499	<0.0001
Not previously diagnosed with depression (n = 233)	0.066	0.942	0.935	1008.4	499	<0.0001

RMSEA = root mean square error approximation

CFI = comparative fit index

TLI = Tucker–Lewis index

df = degrees of freedom

item GF + 5SF solution (Table 3). All coefficient ω s were high, indicating that scale scores adequately captured the blend of general and relevant specific factors. The coefficient ω_H for SF1, SF3, SF4 and SF5 were all low, suggesting that most of the variance in specific factor scale scores for these factors could be attributed to the general factor as against being attributed to the specific factors alone. In contrast, a moderate amount of variance in SF2 scale scores was attributable to that specific factor alone, with the implications considered in the Discussion section.

3.4. Examining for any influence of clinical depression

566 participants answered the question asking if they had been

Table 3

Reliability indices for the 34-item GF + 5SF solutions (when items not bolded in Table 1 were set to zero).

	ω	ω_H
GF	0.96	0.85
SF 1 (Cognitive dysfunction)	0.90	0.39
SF 2 (Empathy loss)	0.82	0.64
SF 3 (Exhaustion)	0.90	0.43
SF 4 (Reduced work performance)	0.88	0.39
SF 5 (Social withdrawal)	0.84	0.36

GF = general factor.

SF = specific factor.

previously diagnosed with depression by a doctor or mental health professional, with 333(58.8%) answering “yes” and 233 (41.2%) answering “no”. Fit statistics for the two subsets are presented in Table 2 and indicate that model fit was satisfactory for both subsets, albeit slightly better in those previously been diagnosed with depression. Further analyses (i.e. the scalar invariance model) identified that those who had not been previously diagnosed with depression had a lower factor score mean on the general factor, as well as on the social withdrawal specific factor (reported in Table 4).

Discussion

We first acknowledge study limitations. Our inclusion criteria were broad, only requiring self-identification of burnout for individuals to participate. We chose this recruitment method as we sought to construct a symptom-based definition of burnout using an inductive or “bottom-up” approach. Our recruitment method does, however, risk some participants having alternate primary conditions misidentified as burnout (e.g. clinical depression, chronic fatigue syndrome). While we did ask participants whether they had ever been previously diagnosed with depression, the reliability of self-report diagnoses can be hindered by several factors, such as the individual’s health literacy (Smith et al., 2008). Further, we did not ask participants who reported having been previously diagnosed with depression which depressive subtypes (e.g., melancholic or non-melancholic depression) they had experienced, a limitation as burnout may overlap with one subtype more than others (see Bianchi et al., 2015b). Thus, the validity of the burnout model we have developed must be examined in future research that incorporates systematic clinical assessment to assess for a primary diagnosis of burnout as against any other condition, as noted by Bianchi et al. (2015b).

Our sample was also predominantly female, and while this gender composition could reflect previous studies suggesting women are more prone than men to experience burnout (e.g., Beauregard et al., 2018; Norlund et al., 2010), it could also reflect women being more likely to self-report psychological distress (Phillips and Segal, 1969). Furthermore, our recruitment of a convenience sample led a predominantly Caucasian and tertiary-educated sample, which is not representative of the general population. As convenience sampling is useful in exploratory research and has been used by other burnout researchers (e.g., de Beer and Bianchi, 2019), the method is defensible, but the generalizability of our results should be interpreted with caution.

Turning to findings, our iterative analyses resulted in a bifactor solution comprising a 34-item general factor and five uncorrelated specific factors. As described, the general factor accounts for the shared variance of all items and captures the target or overarching domain being measured. Thus, we consider our general factor as defining the core symptom set of a burnout syndrome. The high ω_H of the general factor and low ω_H for most of the specific factors suggest that the general factor accounted for the majority of variance in the sample. Rodriguez (2016)

Table 4

Factor score means for the subset not previously diagnosed with depression relative to reference subset with means fixed to zero (values taken from the standardized solution of the scalar invariance model).

	Mean [§]	SE	p [¶]
GF	-0.433	0.095	0.000
SF 1 (Cognitive dysfunction)	-0.070	0.133	0.601
SF 2 (Empathy loss)	0.155	0.098	0.114
SF 3 (Exhaustion)	0.104	0.131	0.424
SF 4 (Reduced work performance)	0.169	0.115	0.141
SF 5 (Social withdrawal)	-0.257	0.093	0.006

GF = general factor.

SF = specific factor.

Means in reference subset are all fixed at zero.

Significant differences based on $p < .05$ are bolded.

stated that “when ω_H (for the general factor) is high ($> .80$), total scores can be considered essentially unidimensional, in the sense that the vast majority of reliable variance is attributable to a single common source” (pp. 224–225). A single scale score based on the general factor is therefore preferable in measures derived from such solutions, with subscales of specific factors unnecessary as they “provide little unique, reliable variance” (p. 225). In this case, a ω_H of 0.85 for the general factor indicated that 85% of the variance of general factor scores was attributed to individual differences on the general factor. Thus, burnout as defined by our the 34-item set may potentially be best measured by a single factor, and not require additional subscales based on specific factors. This finding provides conceptual parsimony and shows the potential for subsequent research to validate the use of a scale score calculated as the simple sum of all 34 items.

This unidimensional conceptualisation of burnout contrasts with existing multi-dimensional models of burnout, including the MBI. Unidimensional models of burnout have been proposed previously (e.g. Pines and Aronson, 1988; Shirom and Melamed, 2006), but such models essentially equate burnout with exhaustion, thus risking making the term ‘burnout’ redundant (Schaufeli and Taris, 2005). Items covering exhaustion were dominant in the general factor we identified, aligning with most existing uni- and multi-dimensional burnout models that position exhaustion as the cardinal feature of burnout (e.g., Kristensen et al., 2005; Maslach et al., 2001; Shirom and Melamed 2006). However, our core burnout domain expanded beyond exhaustion, being also dominated by items capturing cognitive dysfunction (e.g. concentration and memory problems). Cognitive symptoms are not included in the MBI, but their prominence in the current study supports our previous quantitative and qualitative burnout studies (Tavella et al., 2020; Tavella and Parker, 2020a; Tavella and Parker, 2020b), and other studies (Beck et al., 2013; Schaufeli and Taris, 2005; Schaufeli et al., 2019; Sandström et al. 2005), suggesting impaired cognition is one of burnout’s “cardinal symptoms” (Bianchi et al., 2015b).

In our earlier quantitative study (Tavella et al., 2020), the cognitive dysfunction items clustered on a specific factor capturing reduced work efficacy, potentially indicating that compromised work performance in burnout may be a consequence of cognitive dysfunction, rather than a being a key “symptom” of burnout in and of itself. A similar interpretation has been put forward in relation to the MBI, with some arguing that the reduced professional accomplishment subscale measures a consequence of burnout rather than being a key symptom (Bianchi et al., 2018; Kristensen et al., 2005; Schaufeli and Van Dierendonck, 1993). In the current analyses, however, cognitive issues and reduced work accomplishment were captured on separate specific factors, leaving the question as to whether reduced professional achievement is intrinsic to burnout, a consequence of cognitive dysfunction, or a consequence of burnout unresolved. Future longitudinal studies examining the developmental trajectory of cognitive and work performance issues in burnout are needed to resolve this central issue.

Items representing empathy loss were captured in a specific factor, while loading poorly on the general factor. The ω_H index signalled that a moderate amount of variance ($\omega_H = 0.64$) in the loss of empathy factor scores was attributable to this factor’s specific underlying construct, unlike observed scores on the remaining specific factors, with variance in these scores largely attributable to the general (i.e. “total” burnout) factor. This finding suggests while our four other specific factors make up the core aspects of burnout, loss of empathy may be a more independent construct. This independence may reflect loss of empathy and detachment from service recipients being a strategy employed by workers to cope with burnout, rather than a symptom of burnout itself, as suggested by Bianchi et al. (2018). Maslach and colleagues also considered that empathy loss may not be inherent to burnout as experienced by the general population, as when they adapted the MBI-HSS to derive the MBI-GS they replaced the depersonalisation scale with a “cynicism” scale, capturing feelings of detachment from the work environment rather than loss of empathy towards service recipients

(Maslach et al., 2016). The social symptoms of burnout in our sample may be better reflected in the social withdrawal items, such as “I withdraw from family and friends” and “I keep to myself”, which denoted disconnection from personal relationships and reduced enjoyment in socialising. These items loaded higher on the general factor than did the items representing empathy loss, indicating the former to be greater contributors to the core burnout syndrome than the latter.

Of central importance was the presence of depressive symptoms in our final solution, with such items having moderate to high loadings on the general factor. As mentioned in the Introduction, no depression items were apparently included in the item set used in developing the MBI, disallowing their emergence in the final MBI measure even if integral to burnout. As there has been a longstanding debate as to whether depression and burnout are synonymous, we judged inclusion of depression items to be a necessity. We included 31 items taken from established measures of depression in our original 137-item list, but with only four were quantified as contributing to our final solution. One captured low mood, another covered reduced self-worth, and two comprised anhedonia (although these may be more reflective of social withdrawal specifically, as they were picked up in the relevant specific factor). In quality they more suggest depressed mood symptoms rather than clinical depression (where items capturing psychomotor disturbance and suicide are more likely to be prominent). If burnout is synonymous with depression, we would have anticipated multiple depressive symptoms loading highly on the general factor or for analyses to have generated a specific depression factor. As neither occurred, we judge that burnout is unlikely to be synonymous with depression but that a burnout syndrome is commonly if not always accompanied by some depressive symptoms, whether they be part of the syndrome (as reflecting general psychological distress) or a consequence in response to its emergence and impact.

There remains the possibility that some in our sample in fact had a primary diagnosis of clinical depression rather than burnout. Our analysis of measurement invariance suggested that those who reported having been previously diagnosed with depression did have higher “total” burnout (i.e. general factor) scale scores, and higher scores on the social withdrawal specific factor. As reported, the social withdrawal factor contained items indicative of anhedonia, so it is not unexpected that a sub-set of sample members with clinical depression would score higher on this factor and the overall general factor (because of the anhedonia items also being included in general factor scores).

We previously asked a sample of those who reported having had burnout as well as a previous diagnosis of depression whether and how they could differentiate between the two conditions (Tavella and Parker, 2020b). A qualitative analysis of the responses identified numerous judged differences, highlighting that, at least from sufferers’ perspectives, the two conditions can be distinguished. It also needs to be conceded that those with clinical depression are not immune to developing burnout while those with burnout (especially if induced by a toxic work environment) may develop clinical depressive states. While issues of comorbidity and sample ‘purity’ risk merging depression and burnout, we suggest again that our finding of only a partial contribution of depressive items to our final factorial solution (when so many items were evaluated) does not support the view that depression and burnout are synonymous.

Overall, our results suggest a single scale measure of burnout derived from our general factor may be the appropriate strategy for measuring burnout. The utility of single-scale measures derived from bifactor models has been noted by Reise (2012), who stated that “bifactor modeling is one solution to the interpretative mess that often is created when researchers force multidimensional item response data into a unidimensional measurement model.” Our unidimensional model defines burnout as a more heterogeneous condition (particularly in including an impaired cognitive component) than captured by the MBI triadic model, and should be validated against systematic clinical observations of those with burnout in future studies.

Author statement

Gabriela Tavella: Conceptualization; Data curation; Formal analysis; Investigation; Methodology; Project administration; Roles/Writing - original draft; Writing - review & editing.

Dusan Hadzi-Pavlovic: Data curation; Formal analysis; Writing - review & editing.

Gordon Parker: Conceptualization; Funding acquisition; Investigation; Methodology; Project administration; Supervision; Writing - review & editing.

Funding

This work was supported by an Australian Government Research Training Program Scholarship, and by the Australian National Health and Medical Research Council (NHMRC; grant number GNT1176689). The contents of the published material are solely the responsibility of the individual authors and do not reflect the views of the NHMRC.

Acknowledgment

We thank the Black Dog Institute for recruitment assistance.

Declarations of interest

None

References

- Ahola, K., Honkonen, T., Isometsä, E., Kalimo, R., Nykyri, E., Aromaa, A., et al., 2005. The relationship between job-related burnout and depressive disorders—Results from the Finnish Health 2000 Study. *J. Affect. Disord.* 88, 55–62.
- APA, 2013. Diagnostic and statistical manual of mental disorders, 5th Ed. APA, Arlington VA.
- Asparouhov, T. and Muthén, B., 2020. IRT in Mplus. <https://www.statmodel.com/download/MplusIRT.pdf> (accessed 14 January 2021).
- Beauregard, N., Marchand, A., Bilodeau, J., Durand, P., Demers, A., Haines III, V.Y., 2018. Gendered pathways to burnout: results from the SALVEO Study. *Ann. Work Expos. Heal.* 62, 426–437.
- Beck, J., Gerber, M., Brand, S., Puehse, U., Holsboer-Trachsler, E., 2013. Executive function performance is reduced during occupational burnout but can recover to the level of healthy controls. *J. Psychiatr. Res.* 47, 1824–1830.
- Beck, A.T., Ward, C.H., Mendelson, M., Mock, J., Erbaugh, J., 1961. An inventory for measuring depression. *Arch. Gen. Psychiatry.* 4 (6), 561–571.
- Bianchi, R., Boffy, C., Hingray, C., Truchot, D., Laurent, E., 2013. Comparative symptomatology of burnout and depression. *J. Health Psychol.* 18, 782–787.
- Bianchi, R., Schonfeld, I.S., Laurent, E., 2015a. Is it time to consider the "burnout syndrome" a distinct illness? *Front. Public Health.* 3, 158.
- Bianchi, R., Schonfeld, I.S., Laurent, E., 2015b. Burnout–depression overlap: A review. *Clin. Psychol. Rev.* 36, 28–41.
- Bianchi, R., Schonfeld, I.S., Laurent, E., 2018. Burnout Syndrome and Depression. Springer, Singapore, pp. 187–202. In: Y. Kim (Ed.), *Understanding depression*.
- Bianchi, R., Schonfeld, I.S., Laurent, E., 2019. Burnout: moving beyond the status quo. *Int. J. Stress Manag.* 26, 36–45.
- Bowling, A., 2005. Just one question: if one question works why ask several? *J. Epidemiol. Community Health.* 59, 342–345.
- Bretland, R.J., Thorsteinsson, E.B., 2015. Reducing workplace burnout: the relative benefits of cardiovascular and resistance exercise. *PeerJ* 3, e891.
- Brunner, M., Nagy, G., Wilhelm, O., 2012. A tutorial on hierarchically structured constructs. *J. Pers.* 80, 796–846.
- Chen, F.F., West, S.G., Sousa, K.H., 2006. A comparison of bifactor and second-order models of quality of life. *Multivariate Behav. Res.* 41, 189–225.
- Cox, T., Tisserand, M., Taris, T., 2005. The conceptualization and measurement of burnout: questions and directions. *Work Stress* 19, 187–191.
- De Beer, L.T., Bianchi, R., 2019. Confirmatory factor analysis of the Maslach Burnout inventory: a Bayesian structural equation modeling approach. *Eur. J. Psychol. Assess.* 35, 217.
- Dyrbye, L.N., Thomas, M.R., Massie, F.S., Power, D.V., Eacker, A., Harper, W., et al., 2008. Burnout and suicidal ideation among U.S. medical students. *Ann. Intern. Med.* 149, 334–341.
- Garson, G.D., 2015. Structural EQUATION MODELING. G.D.Garson and Statistical Associates Publishing. Asheboro NC.
- Glass, D.C., McKnight, J.D., 1996. Perceived control, depressive symptomatology, and professional burnout: a review of the evidence. *Psychol. Health.* 11, 23–48.
- Hamilton, M., 1960. A rating scale for depression. *J. Neurol. Neurosurg. Psychiatry.* 23, 56–62.
- Heinemann, I.V., Heinemann, T., 2017. Burnout research: emergence and scientific investigation of a contested diagnosis. *SAGE Open.* <https://doi.org/10.1177/2158244017697154>.
- Hu, L.T., Bentler, P.M., 1999. Cutoff criteria for fit indexes in covariance structure analysis: conventional criteria versus new alternatives. *Struct. Equ. Modeling.* 6, 1–55.
- Kristensen, T.S., Borritz, M., Villadsen, E., Christensen, K.B., 2005. The Copenhagen burnout inventory: a new tool for the assessment of burnout. *Work Stress.* 19, 192–207.
- Kroenke, K., Spitzer, R.L., 2002. The PHQ-9: a new depression diagnostic and severity measure. *Psychiatr. Ann.* 32, 509–515.
- Leiter, M.P., Durup, J., 1994. The discriminant validity of burnout and depression: a confirmatory factor analytic study. *Anxiety Stress Coping.* 7, 357–373.
- Maslach, C., Jackson, S.E., 1981. The measurement of experienced burnout. *J. Organ. Behav.* 2, 99–113.
- Maslach, C., Jackson, S.E., Leiter, M.P., 2016. Maslach Burnout Inventory, 4th Ed. Mind Garden Inc, Menlo Park CA.
- Maslach, C., Schaufeli, W.B., Leiter, M.P., 2001. Job burnout. *Annu. Rev. Psychol.* 52, 397–422.
- Mészáros, V., Ádám, S., Szabó, M., Szigeti, R., Urbán, R., 2014. The bifactor model of the Maslach burnout inventory–human services survey (MBI-HSS) - an alternative measurement model of burnout. *Stress Health.* 30, 82–8.
- Muthén, L.K., Muthén, B.O., 2002. Mplus VERSION 2.12: Addendum to the Mplus user's guide. Muthén and Muthén. Los Angeles CA.
- Norlund, S., Reuterwall, C., Höög, J., Lindahl, B., Janlert, U., Birgander, L.S., 2010. Burnout, working conditions and gender-results from the northern Sweden MONICA Study. *BMC Public Health* 10, 326. <https://doi.org/10.1186/1471-2458-10-326>.
- Parker, G., McCraw, S., Blanch, B., Hadzi-Pavlovic, D., Synnott, H., Rees, A.M., 2013. Discriminating melancholic and non-melancholic depression by prototypic clinical features. *J. Affect. Disord.* 144, 199–207.
- Phillips, D.L., Segal, B.E., 1969. Sexual status and psychiatric symptoms. *Am. Sociol. Rev.* 34, 58–72.
- Pines, A.M., Aronson, E., Kafry, D., 1988. Burnout: From Tedium to Personal Growth. Career Burnout. Free Press, New York.
- Reise, S.P., 2012. The rediscovery of bifactor measurement models. *Multivariate Behav. Res.* 47, 667–696.
- Reise, S.P., Kim, D.S., Mansolf, M., Widaman, K.F., 2016. Is the bifactor model a better model or is it just better at modeling implausible responses? Application of iteratively reweighted least squares to the Rosenberg Self-Esteem Scale. *Multivariate Behav. Res.* 51, 818–838.
- Rodriguez, A., Reise, S.P., Haviland, M.G., 2016. Applying bifactor statistical indices in the evaluation of psychological measures. *J. Pers. Assess.* 98, 223–237.
- Sandström, A., Rhodin, I.N., Lundberg, M., Olsson, T., Nyberg, L., 2005. Impaired cognitive performance in patients with chronic burnout syndrome. *Biol. Psychol.* 69, 271–279.
- Schaufeli, W.B., 2003. Past performance and future perspectives of burnout research. *SA J. Ind. Psychol.* 29, 1–15.
- Schaufeli, W.B., De Witte, H., Desart, S., 2019. Manual Burnout Assessment Tool (BAT): Unpublished Internal Report. KU Leuven, Belgium.
- Schaufeli, W.B., Leiter, M.P., Maslach, C., 2009. Burnout: 35 years of research and practice. *Career Dev. Int.* 14, 204–220.
- Schaufeli, W.B., Taris, T.W., 2005. The conceptualization and measurement of burnout: common ground and worlds apart. *Work Stress* 19, 256–262.
- Schaufeli, W.B., Van Dierendonck, D., 1993. The construct validity of two burnout measures. *J. Org. Behav.* 14, 631–647.
- Shirom, A., Melamed, S., 2006. A comparison of the construct validity of two burnout measures in two groups of professionals. *Int. J. Stress Manag.* 13, 176–200.
- Smith, B., Chu, L.K., Smith, T.C., Amoroso, P.J., Boyko, E.J., Hooper, T.I., et al., 2008. Challenges of self-reported medical conditions and electronic medical records among members of a large military cohort. *BMC Med. Res. Methodol.* 8, 37. <https://doi.org/10.1186/1471-2288-8-37>.
- Svanborg, P., Åsberg, M., 1994. A new self-rating scale for depression and anxiety states based on the comprehensive psychopathological rating scale. *Acta Psychiatr. Scand.* 89, 21–28.
- Tavella, G., Parker, G., 2020a. A qualitative reexamination of the key features of burnout. *J. Nerv. Ment. Dis.* 208, 452–458.
- Tavella, G., Parker, G., 2020b. Distinguishing burnout from depression: An exploratory qualitative study. *Psychiatry Res* 291, 113212. <https://doi.org/10.1016/j.psychres.2020.113212>.
- Tavella, G., Hadzi-Pavlovic, D., Parker, G., 2020. Burnout: re-examining its key constructs. *Psychiatry Res* 287, 112917. <https://doi.org/10.1016/j.psychres.2020.112917>.
- Wan, T.T.H., 2002. Evidence-Based Health Care Management: Multivariate Modeling Approaches, 1st Ed. Kluwer Academic Publishers, Norwell, MA.
- WHO, 2018. International Statistical Classification of Diseases and Related Health Problems (11th revision). WHO, Geneva, Switzerland.
- WHO, 2019. Burn-out an "occupational phenomenon": international classification of diseases. <https://www.who.int/news/item/28-05-2019-burn-out-an-occupational-phenomenon-international-classification-of-diseases> (accessed 10 February 2021).
- Worley, J.A., Vassar, M., Wheeler, D.L., Barnes, L.L., 2008. Factor structure of scores from the Maslach Burnout inventory: a review and meta-analysis of 45 exploratory and confirmatory factor-analytic studies. *Educ. Psychol. Meas.* 68, 797–823.